

● HARD

● ADVANCED MATH

Advanced Math

13

10 questions · ~15 min

Q1

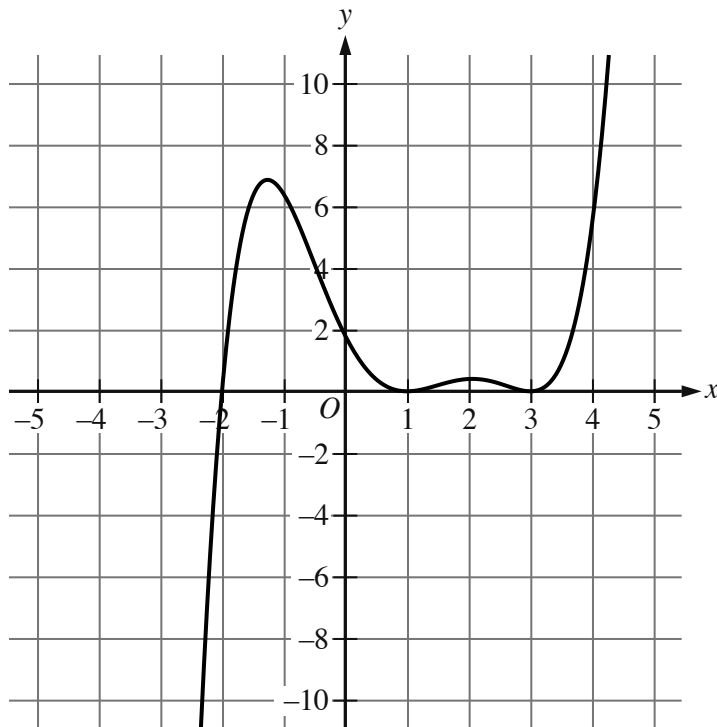
GRID-IN ADVANCED MATH

In the xy -plane, the graph of $y = 3x^2 - 14x$ intersects the graph of $y = x$ at the points $(0, 0)$ and (a, a) . What is the value of a ?

STUDENT-PRODUCED RESPONSE

.	/	.	.
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.



- Moving from left to right, the curve passes from quadrant 3 to quadrant 2 to quadrant 1.
- The curve has 2 relative maximums and 2 relative minimums.
- The curve crosses the x axis at the point (negative 2 comma 0).
- The curve touches the x axis at the following points:
 - (1 comma 0)
 - (3 comma 0)

The graph of $y = gx$ is shown. If $gx + 28 = \frac{1}{10}x - a^2x - b^2x - c$, where a , b , and c are constants, which of the following could be the value of b ?

- A. 31
- B. 3
- C. -2

D. -25

Q3

GRID-IN

ADVANCED MATH

$$x - 47^2 = 1$$

What is the sum of the solutions to the given equation?

STUDENT-PRODUCED RESPONSE

.	/	.	.
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.

Q4

GRID-IN

ADVANCED MATH

$$f(x) = x^2 - 48x + 2,304$$

What is the minimum value of the given function?

STUDENT-PRODUCED RESPONSE

.	/	.	.
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.

Q5

ADVANCED MATH

$$y = 2x^2 - 21x + 64$$

$$y = 3x + a$$

In the given system of equations, a is a constant. The graphs of the equations in the given system intersect at exactly one point, (x, y) , in the xy -plane. What is the value of x ?

- A. -8
- B. -6
- C. 6
- D. 8

Q6

ADVANCED MATH

The function h is defined by $h(x) = a^x + b$, where a and b are positive constants. The graph of $y = h(x)$ in the xy -plane passes through the points $(0, 10)$ and $(-2, \frac{325}{36})$.

What is the value of ab ?

- A. $\frac{1}{4}$
- B. $\frac{1}{2}$
- C. 54
- D. 60

Q7

ADVANCED MATH

In the xy -plane, the graph of $y = x^2 - 9$ intersects line p at $(1, a)$ and $(5, b)$, where a and b are constants. What is the slope of line p ?

- A. 6
- B. 2
- C. -2
- D. -6

Q8

ADVANCED MATH

$$f(x) = 9(4)^x$$

The function f is defined by the given equation. If $g(x) = f(x + 2)$, which of the following equations defines the function g ?

- A. $g(x) = 184^x$
- B. $g(x) = 1444^x$
- C. $g(x) = 188^x$
- D. $g(x) = 8116^x$

Q9

ADVANCED MATH

$$\frac{20}{p} = \frac{20}{q} - \frac{20}{r} - \frac{20}{s}$$

The given equation relates the positive variables p , q , r , and s . Which of the following is equivalent to q ?

- A. $p + r + s$
- B. $20p + r + s$
- C. $\frac{prs}{pr + ps + rs}$
- D. $\frac{prs}{20p + 20r + 20s}$

Q10

ADVANCED MATH

$$fx = -6x^2 + 60x - 126$$

The function f is defined by the given equation. Which of the following equivalent forms of the equation displays the maximum value of the function as a constant or coefficient?

- A. $fx = -6x^2 + 42x + 18x - 126$
- B. $fx = -6x(x - 7) + 18(x - 7)$
- C. $fx = -6x - 5^2 + 24$
- D. $fx = -6x - 7x - 3$